Real-Time Sign Language Recognition

Elevate Labs AI & ML Internship (MSME, Govt. of India) – Project Phase | Ayesha Nusrath

1. Introduction

Sign language is the main way many deaf and hard-of-hearing people communicate, but most people do not understand it. A system that converts hand gestures to text in real time can reduce this communication gap. This project recognises the static letters of the American Sign Language (ASL) alphabet from a camera image and joins the recognised letters into words.

2. Abstract

The system uses MediaPipe to locate 21 landmarks on the hand, converts them into a 63-value feature vector that is independent of hand position and size, and classifies that vector with a small Keras neural network into one of 28 classes (A–Z, *space*, *del*). Trained on 10,626 landmark samples from the Kaggle ASL Alphabet dataset, the model reaches **97.55% accuracy on a held-out test set** (96.06% on training data). A webcam script applies the model to live video, smooths predictions over several frames, and types a letter once it has been held steady, so that words can be spelled on screen.

3. Tools Used

Python	Programming language
MediaPipe (HandLandmarker)	Hand detection and 21-point landmark extraction
TensorFlow / Keras	Building and training the neural network
OpenCV	Webcam capture, drawing and on-screen display
NumPy, scikit-learn, Matplotlib	Feature handling, train/test split, metrics, graphs
Google Colab, Kaggle	Free cloud notebook for training; ASL Alphabet dataset

4. Steps Involved in Building the Project

- **Dataset:** downloaded the ASL Alphabet dataset (3000 images per class, 29 classes). The *nothing* class was dropped because “no hand detected” already means no sign in live use. 500 random images per class were used.
- **Hand landmarks:** MediaPipe’s HandLandmarker (Tasks API) returned 21 points (x, y, z) per hand. A hand was found in 10,626 of the 14,000 images; the rest were skipped.
- **Feature preparation:** points were shifted so the wrist is the origin and divided by the largest distance, giving 63 normalised numbers per image.
- **Train/test split:** 80% training, 20% testing, stratified so every sign appears in both parts.
- **Model:** Dense(128, ReLU) → Dropout(0.3) → Dense(64, ReLU) → Dropout(0.3) → Dense(28, Softmax); Adam optimiser, sparse categorical cross-entropy, 30 epochs, batch size 32.
- **Real-time recognition:** each webcam frame goes through the same landmark and normalisation steps. Predictions below 80% confidence are ignored, a majority vote over the last 8 frames removes flicker, and a sign held for about 1.2 seconds is typed. *space* adds a space and *del* removes the last character.

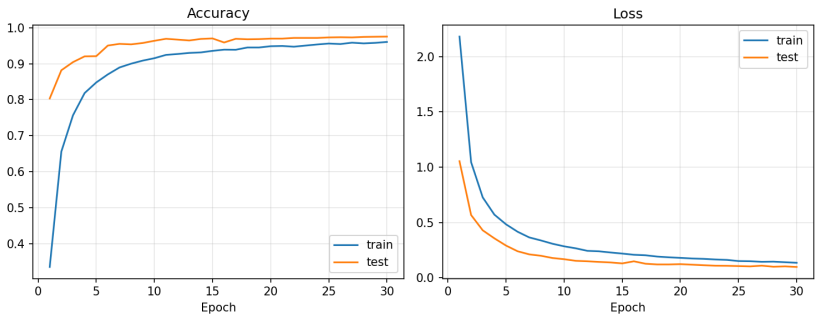


Figure 1: Accuracy and loss over 30 epochs.

5. Conclusion

Using hand landmarks instead of raw pixels gave a lightweight classifier that trains in about a minute and reaches 97.55% test accuracy on 28 classes. The accuracy figure comes from the dataset’s own test split, so performance with other signers, cameras or lighting may be lower. Letters that need motion (J and Z) are handled as single frames, and only one hand is tracked. Future work: collect extra images from more signers, add motion-based signs, and add text-to-speech output.